

CSC 2202 Data Communication and Computer Networks

COURSE INFORMATION SHEET

PROGRAMME: Computer Science	DEGREE: BS
COURSE: Data Communication and Computer Networks	SEMESTER: Spring 2023
COURSECODE: CSC 3205	COURSE TYPE: CORE
COURSE AREA/DOMAIN: Computer Science	CONTACTHOURS: 3 hours/Week
CORRESPONDINGLAB COURSECODE(IFANY): Yes	LAB COURSENAME: Lab- Data Communication and Computer Networks

COURSE OBJECTIVES:

By the end of this course, students will be able to: 1. Build an understanding of the fundamental concepts of computer networking. 2. Familiarize the student with the basic taxonomy and terminology of the computer networking area. 3. Introduce the student to advanced networking concepts, preparing the students for entry Advanced courses in computer networking. 4. Allow the student to gain expertise in some specific areas of networking such as the design and maintenance of individual networks.

COURSE DESCRIPTION:

This course provides the students with an overview of the concepts and fundamentals of data communication and computer networks. Topics to be covered include : data communication concepts and techniques in a layered network architecture, communications switching and routing, types of communication, network congestion, network topologies, network configuration and management, network model components, layered network models(OSI reference model, TCP/IP networking architecture) and their protocols, various types of networks (LAN,MAN,WAN and Wireless networks) and their protocols.

SYLLABUS:

Week	Topics
1.	Introduction to communication and networks, internet, network edge. Importance of protocols and algorithms to provide seamless communication. Necessity of layered architecture for globalization in communication networks.
2.	OSI/ TCP/IP layers, Network core, circuit switching and packet switching network performance metrics : delay, path loss, throughput
3.	Network types, LAN MAN WAN, latest networks i.e. low power WAN. Future of networks- IoT. communication medium and types of medium (wired wireless)
4.	Application layer : principles and protocols http, FTP, SMTP, DNS, peer to peer applications. Socket programming concepts.

5.	Transport layer, Multiplexing and de multiplexing transport layer protocols TCP, UDP (connection less and connection oriented protocols).
6.	The network layer: devices of network layer. Internet protocol, routing in internet
7.	Network Layer (Layer 3) wireless protocols for adhoc and sensor networks, Addressing, Sub netting
8.	Mid Term Paper.
9.	Introduction to link layer, MAC and LLC, Medium access techniques, link virtualization, data center networking.
10.	Medium access techniques ALOHA, Slotted Aloha, CSMA/CD, CSMA/ CA
11.	Medium Access Control protocols- LLC protocols ARP, RARP
12.	Wireless and mobile networks, an introduction of IEEE 802.11, cellular network, mobile IP, IEEE 802.15.4 and IEEE 802.15.6
13.	Multimedia networking, streaming and stored video, Voice over IP
14.	Network security fundamentals, cryptography
15.	Network management. Infrastructure and internet standard management framework. Ubiquitous networks- combination of AI and networks.

TEXT/REFERENCE BOOKS:

Computer Networking: A Top-Down Approach Featuring the Internet, 6th edition by James F. Kurose and Keith W. Ross , James F. Kurose and Keith W. Ross , null Edition

Data and Computer Communications, 10th Edition by William Stallings , William Stallings, 1st Edition, ,2018.

Data Communication and Computer Networks, 5th Edition by Behrouz A. Forouzan, Behrouz A. Forouzan

COURSE LEARNING OUTCOMES (CLOs):

CLO	DESCRIPTION	Domain	BT Level
1	Describe the key terminologies and technologies of computer networks	b	1
2	Explain the services and functions provided by each layer in the Internet protocol stack.	c	2
3	Identify various internetworking devices and protocols, and their functions in a network.	c	2
4	Analyze working and performance of key technologies, algorithms and protocols	c	2
5	Build Computer Network on various Topologies	d	3